A learner-free, self-organizing swarm approaches reinforcement learning's explore–exploit optimum

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Abstract

The explore–exploit trade-off is the core problem of reinforcement learning (RL). We show that a minimal foraging swarm *approaches* this optimum's allocation—the ideal-free / Nash equilibrium of a multi-agent resource-sharing game (an equilibrium allocation, not a return maximum)—with no learner, no trained parameters, and no reward signal in its dynamics. The reinforcement learning is at the level of the *problem* solved and the *allocation reached*, not the micro-rule. Agents explore by heavy-tailed (Lévy) flights, recruit neighbours to a discovery through a threshold cascade on their proximity network, and disperse as a patch crowds; an ablation shows this coupling of recruitment to movement is what drives foragers into proportion to resource richness. The swarm reaches the allocation rather than learning it, and attains the *shape* of the optimum but not its exact equilibrium—it stays measurably exploitable (Nash-gap 0.58, versus ≈0 for a converged learner). Against PPO learners trained in the same sensing-limited environment, the reward-free swarm is competitive at zero training: its harvest ties the best learner where resource is clustered and plentiful (both ≈0.80 of a validated oracle ceiling) and trails only where resource is scarce, while it matches the ideal-free allocation gradient more faithfully than any learner—learning buys precision, not a better allocation. Neither learned message passing nor a centralized critic overturns this. We read it not as beating RL, but as evidence that a minimal swarm's self-organizing physics reaches the optimum RL seeks with none of RL's machinery. A companion paper develops the mechanism.

1 Introduction

Reinforcement learning (RL) is the machine-learning account of the explore–exploit trade-off: a system must *explore* to discover where reward is and *exploit* what it has found, and good behaviour is the behaviour that balances the two and settles at the best available payoff (Sutton & Barto, 2018). A foraging swarm faces exactly this problem. Patches of resource are options of unknown location and yield; the swarm must search for them and converge on the good ones, with only local sensing and no central controller. Because many foragers share each patch, a forager's best available payoff is congestion-limited—the most it can get given where everyone else is—so the problem is multi-agent and its optimum is a collective *allocation* (each forager at its best available payoff: the ideal-free / Nash equilibrium), not a single best option. The natural question is whether a swarm that forages well is, in effect, reaching the optimum of this reinforcement-learning problem, and if so, what plays the part of the learner.

We answer: the swarm approaches the optimum—it attains the optimal allocation's *shape*—and there is no learner, and, more strongly, no reward signal inside the dynamics at all. A minimal self-organizing mechanism—Lévy exploration, threshold recruitment to discoveries, and density-dependent dispersal—balances exploration and exploitation and converges toward the optimal allocation of foragers across patches (reaching its shape, though it remains measurably exploitable; §4, §6). The point of this paper is a claim about *level of abstraction*. The reinforcement learning is at the level of the *problem solved* and the *outcome reached*, not at the level of the micro-rule. We are explicit about this because a tempting reading—that the recruitment rule is itself a reward-following learning update—is false, and we show it is false. The agents never evaluate how good a patch is; recruitment responds to local crowding, not to payoff. What is remark-

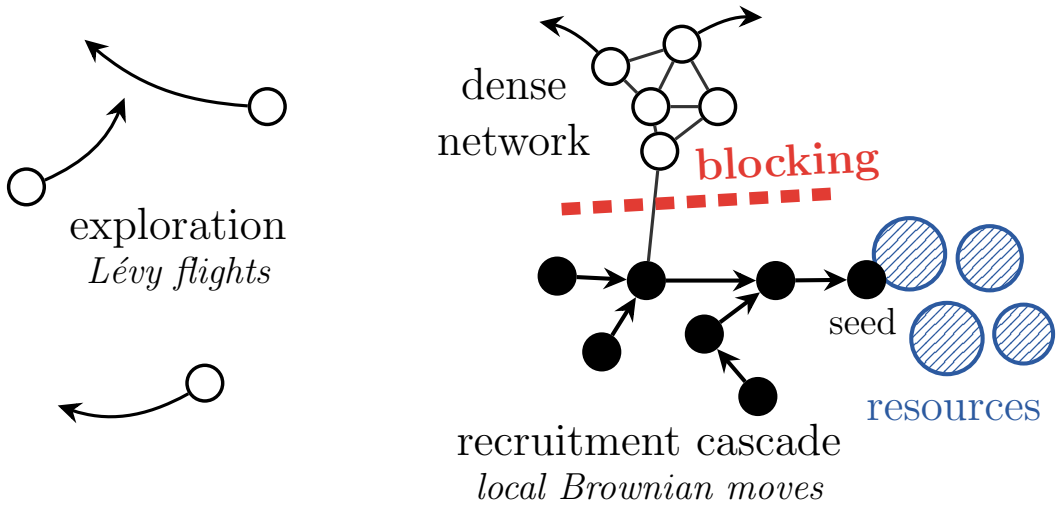


Figure 1: **The physical model.** Agents explore by Lévy flights (left). An agent that lands on a resource—a *seed*—triggers a recruitment cascade along the proximity network, drawing neighbours into local convergence on the find; as the neighbourhood crowds, degree-normalized activation a/k falls below threshold ϕ and further recruitment is *blocked* (red). Schematic of the model described in §3; the full model is developed in (Wilkerson, 2026).

able is precisely that a system with *none* of reinforcement learning's machinery—no value function, no policy parameters, no training, no reward signal—nonetheless lands near the optimum reinforcement learning is built to find.

Claims. The paper makes four claims, each stated so it can be checked against evidence.

1. **The swarm approaches the explore–exploit optimum.** The self-organizing cascade cycle balances exploration and exploitation and converges to the *shape* of the optimal foraging allocation—the ideal free distribution (matching slope 0.94)—reached dynamically and caused by the recruitment–movement coupling (remove it and matching collapses to 0.08); the exact equilibrium is not reached—the swarm stays measurably exploitable (§4, §6).
2. **Learner-free and reward-free.** It does so with no value function, no policy parameters, and no training—and the dynamics contain no reward signal: the agents never read a payoff, and recruitment is driven by crowding, not by reward (§5).
3. **It reaches the allocation, it does not learn.** Reaching this allocation is not epistemic learning: the configuration's informativeness rises once and plateaus, and re-settles after the resources move with no cumulative gain (§5).
4. **Competitive with trained PPO/MAPPO/GNN baselines under matched embodiment, and the most gradient-faithful in allocation.** With sensing primitives matched so neither arm can approach unsensed resources, the reward-free swarm at *zero* training is *statistically equivalent* to the best trained PPO learner's harvest in a clustered, resource-rich field ($\approx$0.80 vs $\approx$0.80 of a validated oracle ceiling; equivalent within a $\pm$0.02 margin by a two-one-sided-test, 90% CI of the difference $[-0.004, +0.008]$) and matches the ideal-free allocation gradient most faithfully of the arms tested in both fields (slope $\approx$1 vs the learners' 0.69–0.91, i.e. unbiased where they under-concentrate)—including the scarce field where the learner collects more total intake ($\approx$0.33 vs $\approx$0.22). Learning buys *precision* (a tighter per-capita intake), not a better gradient match; and in the clustered regime neither learned message passing over the same network nor a centralized-critic (CTDE) learner beats it (§6). The swarm does not, however, transfer to a different reward structure—a fixed, reward-blind mechanism cannot.

What is, and is not, claimed. We claim that the *optimum* of a reinforcement-learning problem—the optimal explore–exploit allocation, the ideal-free / Nash equilibrium—is reached by a physical, self-organizing mechanism with no learner. We do *not* claim that the micro-dynamics are a reinforcement-learning algorithm, a reward-following update, or the replicator equation: the recruitment rule is reward-free *by construction* (no step reads payoff; §3, §5). We do *not* claim the optimum is a total-reward maximum—under resource sharing it is an equal-intake equilibrium (App. B). And we do *not* claim epistemic learning: the swarm reaches the optimum, it does not improve an estimate from experience. Return genuinely varies—and the explore–exploit stakes bite—only in the *dynamic, embodied* setting (§6), where cumulative intake differs across policies; that is where we compare to a learner. The content is both precise and, we think, important: a system with *none* of reinforcement learning's machinery lands exactly where learning-in-games dynamics converge, and under matched embodiment is competitive with a trained learner—statistically tying its harvest in resource-rich fields and matching the allocation gradient most faithfully everywhere, while a trained learner collects more raw intake only where resource is scarce and buys lower per-capita variance (§6). We do *not* claim the swarm out-harvests a trained learner universally, nor that its allocation is the most precise; we claim it reaches the *shape* of the problem's optimum—the ideal-free allocation, unbiased with respect to the resource gradient—at least as faithfully as any learner, with no learner, reward, or training. How the harvest comparison comes out depends on the resource regime and the primitives the arms share, a caution for how such comparisons are run.

Scope and the companion paper. The swarm's mechanism—a spatial threshold recruitment cascade whose self-organized "cascade window" is analysed in full—is developed in a companion paper (Wilkerson, 2026), which we cite for the complete treatment. This paper is *self-contained*: every claim it makes is supported by evidence shown here. In particular, the self-organizing operating regime is demonstrated in-paper (§4, Fig. 5), the recruit-state movement property we use is a stated feature of the model (below), and the learner comparison matches movement by construction (App. E)—so no load-bearing claim depends on the companion. We summarise the mechanism only as far as the present reading needs; our claims concern the explore–exploit problem and the allocation the swarm reaches.

2 Related work

The explore–exploit problem and bandits. The multi-armed bandit (Robbins, 1952; Lattimore & Szepesvári, 2020) is the canonical formalisation of explore–exploit, and the one-state reduction of the general RL problem (Sutton & Barto, 2018). Our setting is its multi-agent, resource-sharing instance: arms are patches whose per-capita yield falls as more foragers work them, so the optimum is an allocation, not a single best arm.

The optimum: ideal free distribution and habitat games. The optimal allocation of foragers over patches is the *ideal free distribution* (IFD): foragers distribute in proportion to resource so that per-capita intake is equal and no forager gains by moving (Fretwell & Lucas, 1969). This is the Nash equilibrium of the habitat-selection (congestion) game (Rosenthal, 1973; Wardrop, 1952) and the standard optimality criterion in behavioural ecology. We use the IFD only to *characterise the solution* the swarm reaches; we do not claim the swarm reaches it by playing the game or by any game-solving procedure.

Learning in games as the mean field of RL. Reward-proportional reinforcement is, in the large-population mean field, the replicator equation (Börgers & Sarin, 1997), the bridge used to analyse multi-agent learning dynamically (Hofbauer & Sigmund, 1998; Sandholm, 2010; Tuyls & Nowé, 2005; Bloembergen et al., 2015). This is the level at which one *might* hope to identify a physical swarm with RL. We deliberately do *not* rest our claim there: §5 shows the swarm's micro-dynamics are reward-free and are not a replicator update. Our identification is at the problem/solution level instead.

Deep reinforcement learning for foraging. Trained deep-RL agents can *learn* to forage near-optimally: Wispinski et al. (2023) show policy-gradient agents that approach the marginal-value-theorem patch-leaving optimum and recover hallmarks of animal foraging, a close precedent for bridging RL and foraging behaviour. That work and ours are complementary opposites. There, a *learner* approaches a *single-agent* foraging

Symbol	Meaning	Value
N	foragers	200
L	torus side length	800
$\rho_r = N(r/L)^2$	connectivity density	2.1 ($r \approx 82$)
ϕ	activation threshold ($a/k \geq \phi$)	0.18 ($2/\phi \approx 11$)
α	Lévy exponent	1.0
C	patches (resource clumps)	10
β	Pareto patch-richness exponent	2.2
M	resource sources	80
z_c^{RGG}	percolation onset	$\approx$4.51

Table 1: **Main model parameters** (spatial foraging model, §3). The abstract congestion game (§6, App. D) uses K=10 patches and N=200 agents with the same Pareto richness; the embodied comparison (App. E) uses N=200 over horizon T=600.

optimum by training on reward; here, a *learner-free*, reward-free swarm reaches a *multi-agent* allocation optimum—the ideal-free / congestion-game Nash—with no training and no reward. Setting the two side by side isolates what the learning buys and what the self-organization supplies.

Swarm optimisation, stigmergy, and designed game-solvers. Metaheuristics such as particle-swarm (Kennedy & Eberhart, 1995) and ant-colony optimisation (Dorigo et al., 1996) coordinate many simple units, but they are *designed* to optimise a supplied objective and carry memory (a personal/global best; a pheromone table). Stigmergic foraging is usually read as *reinforcement*, with the pheromone field an external memory / value estimate (Vellinger et al., 2025). Decentralised multi-agent control reaches Nash / potential-game equilibria from local information *by design* (Marden et al., 2009), and bee-inspired allocation reaches a Nash, optimal allocation by designed local rules (Quijano & Passino, 2010). Ours is none of these: it is not a designed optimiser, it carries no memory or value estimate, and it uses no reward—it is a physical mechanism whose self-organized dynamics happen to settle at the problem's optimum. That contrast makes the claim precise: the swarm is the memory-free, reward-free limit these methods are not.

Minimally intelligent agents reaching an optimum. That non-optimising agents can reach a market's competitive allocation is the "zero-intelligence" result of experimental economics (Gode & Sunder, 1993). Our result is in this spirit—the structure of the interaction, not the sophistication of the agent, does the optimising—extended to a spatial, self-organizing foraging swarm and to the explore–exploit problem. Reaching an equilibrium without computing payoffs also connects to no-regret and evolutionary learning dynamics, which provably converge to equilibria of the underlying game (Sandholm, 2010; Hofbauer & Sigmund, 1998); our mechanism is the limiting case that uses no regret signal and no payoff at all. We make no entropy-regularised ("soft") RL claim (Haarnoja et al., 2018; Ziebart et al., 2008): the spread of the converged allocation is the ideal free distribution set by resource sharing, not a temperature.

3 The physical model: a self-organizing forager

The full model, its critical "cascade window," and the efficiency analysis are in the companion paper (Wilkerson, 2026); here is what the present reading needs, drawn in Figs. 1 and 2. N foragers move on a torus. An inactive agent *explores* by heavy-tailed (Lévy) hops. An agent that lands on resource becomes *active* (a *seed*) and *recruits* its neighbours: recruitment is a linear-threshold cascade on the proximity graph (an edge joins agents within sensing range r). An agent activates when the active fraction of its neighbours crosses a threshold, $a/k \geq \phi$ (a active neighbours, k neighbours), and recruited agents switch to short local moves toward the find. Because activation is *degree-normalized*, the same rule that spreads a cascade also stifles it once a neighbourhood crowds and a/k falls back below ϕ: recruitment to a crowded patch shuts off. So the swarm aggregates on a find and then releases to explore again. Crucially for what follows, **no step of this rule reads a payoff**: a seed fires on *contact* with resource, and recruitment responds to the *active fraction* of neighbours—a crowd signal—never to how rich a patch is or how much intake an agent receives.

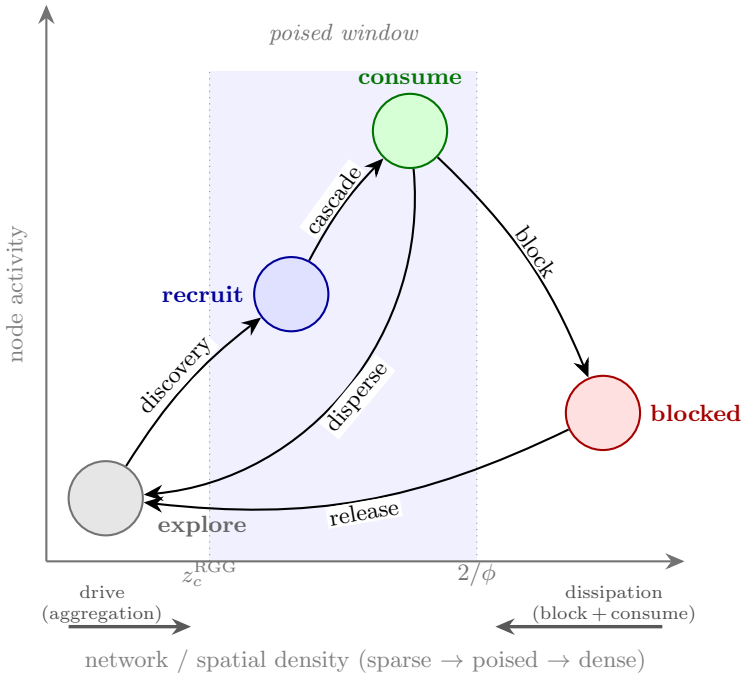


Figure 2: **The self-organizing cycle balances exploration and exploitation.** In the plane of network/spatial density (horizontal) and node activity (vertical), the swarm runs a drive–avalanche cycle: discovery seeds a recruitment cascade (explore→recruit→exploit); crowding blocks and disperses it (exploit→released→explore); aggregation (*drive*) versus dispersal (*dissipation*) hold the swarm in a poised window, with no parameter tuned. Schematic of the self-organizing cycle described in §3; full treatment in (Wilkerson, 2026).

The cascade cycle is an explore–exploit balancer. The mechanism runs a drive–avalanche cycle (Fig. 2): discovery seeds a cascade (explore → recruit → exploit), crowding stifles and disperses it (exploit → released → explore), and the swarm self-organizes to a poised operating point—a "cascade window" in connectivity—without any parameter being tuned. We show this self-tuned operating regime directly in this paper (§4, Fig. 5): the swarm holds a subcritical, near-ideal-matching operating point across a 60× density range with no per-condition tuning. The window's lower edge is the percolation onset and its upper edge the degree-dependent blocking bound; the companion paper maps it in full with phase diagrams over connectivity and activation threshold (Wilkerson, 2026). The density-dependent dispersal ("blocking") is thus what regulates the explore–exploit *balance*—how much of the swarm explores versus exploits. At any instant part of the swarm is exploring and part is exploiting; the cycle continually re-apportions the two—the explore–exploit trade-off managed, in real time, by the physics rather than by a controller. One further property of the model is used below: recruitment is *movement-neutral*—by definition of the recruit state, recruited agents reuse the exploring step length and change only direction, so no movement budget is spent to coordinate (a stated feature of the model here; analysed in (Wilkerson, 2026)).

4 The swarm approaches the explore–exploit optimum

The problem and its optimum. Coarse-grain the swarm by the fraction n_i of agents committed to patch i (the rest exploring); $\mathbf{n}$ is the swarm's allocation over patches. With resource Q_i shared among the Nn_i foragers working patch i, per-capita intake is $f_i = Q_i/(Nn_i)$, which falls as a patch crowds. The optimal allocation—where no forager can raise its intake by moving—equalises per-capita intake across occupied patches, giving $n_i \propto Q_i$: the ideal free distribution, and the Nash equilibrium of the habitat-sharing game (Fretwell & Lucas, 1969; Rosenthal, 1973) (App. B). This is the optimum of the swarm's

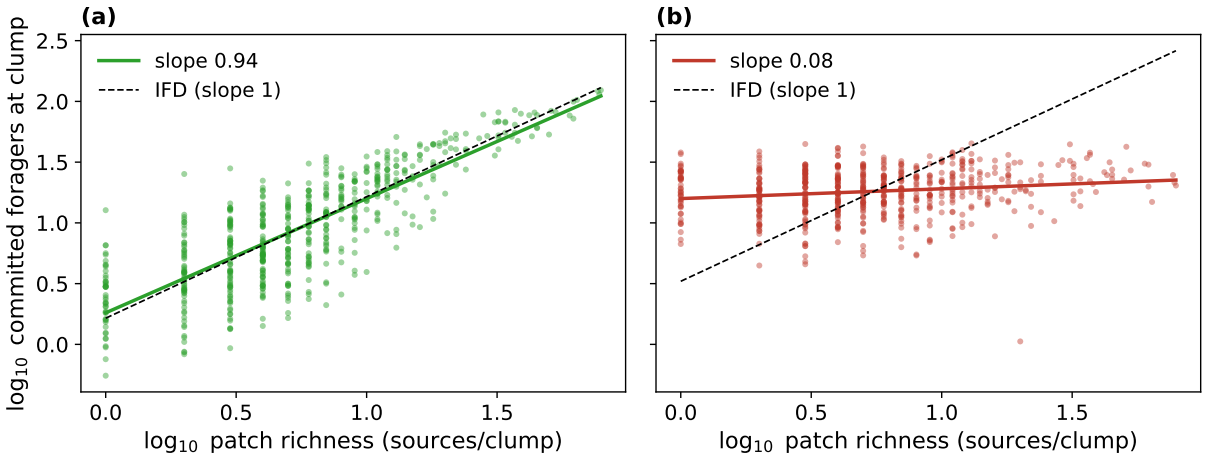


Figure 3: **The swarm reaches the optimal allocation, and the coupling causes it.** Committed foragers per patch vs. richness, pooled over B=64 independent replica worlds (n=591 patch-replica points; the B=64 worlds are the independent units). **(a) Recruiting.** Pooled slope 0.94 (R^2=0.67; the per-replica slope distribution has mean ≈1.0, so the pooled fit is not an artefact of pooling non-independent points); dashed is the ideal slope 1, the small shortfall the "undermatching" real foragers show. **(b) Coupling off.** With recruitment removed, matching is destroyed (slope 0.08): the optimal (ideal-free) allocation is caused by the recruitment–movement coupling.

multi-agent explore–exploit problem: the allocation at which every forager is at its best available payoff. Because resource is shared, total intake $\sum_i n_i f_i = \sum_i Q_i/N$ is the same for any interior allocation (App. B), so this optimum is a load-balancing *equilibrium*, not a total-harvest maximum; the harvesting-efficiency side of foraging—that finding and exploiting resources quickly collects more over time—is the companion paper's result (Wilkerson, 2026).

The swarm reaches it, and the coupling causes it. If the swarm settles at the ideal free distribution, committed foragers per patch should track patch richness with unit slope. In a heterogeneous field (Pareto patch richness over a ∼40-fold range—heavy-tailed, sparse, and clustered, as natural resource fields typically are) the measured slope is 0.94 ($R^2 = 0.67$; Fig. 3a)—near-ideal, with the mild *undermatching* real foragers universally show. This is the *shape* of the optimum (foragers in proportion to richness); how far the swarm sits from the *exact* equilibrium—its exploitability—is a separate question we quantify against a trained learner in §6, where it is measurably exploitable, not at the Nash point. The decisive result for the causal claim is the ablation: *removing the recruitment coupling collapses the slope to* 0.08 (Fig. 3b). The allocation shape is not a property of the individual agents—it is produced by coupling recruitment to movement. (We attribute no finer mechanism to the coupling than this: in particular we do *not* claim the coupling implements a particular learning update—§5.)

A second ablation localizes the mechanism further. Removing the crowding cutoff—the degree-normalized blocking that shuts recruitment off as a patch fills—leaves the matching slope essentially unchanged (0.94 → 0.92, no over-aggregation; App. C). The two ablations together separate the two roles: the allocation *shape* (slope-one matching) is set by richness-proportional discovery and recruitment—it survives removing the cutoff but collapses without the coupling—while the crowding cutoff instead regulates *where* on the explore–exploit balance the swarm operates (the self-organized operating point below), not the shape it reaches.

The allocation is reached dynamically. The matching is not imposed; it is *reached*. Tracking the matching slope through a run (Fig. 4), the coupled swarm climbs from near zero to its ideal-free-shape plateau within ∼150 steps and holds; the uncoupled swarm stays flat. The allocation is an *attractor* of the explore–exploit dynamics—reached quickly and then maintained.

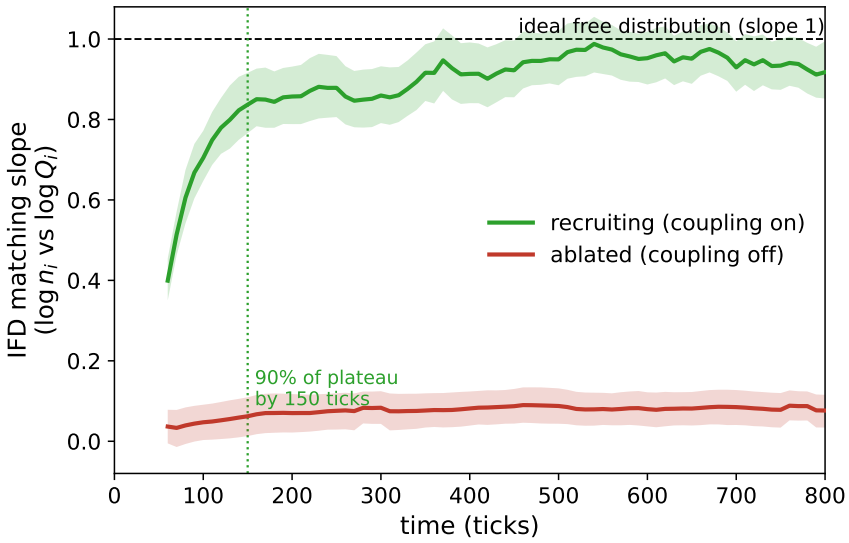


Figure 4: **The allocation is reached dynamically.** Matching slope (regression of log committed foragers per patch on log richness, trailing window; replicas with bootstrap 95% CI) vs. time. With the recruitment coupling (green) the swarm climbs to the ideal-free plateau (slope $\to 1$, dashed) within $\sim$150 steps and holds; without it (red) matching never develops.

The operating regime is self-organized, not tuned. The swarm is not placed at a hand-chosen operating point. Sweeping connectivity ρ_r over a 60$\times$ density range (Fig. 5), the swarm holds a *subcritical* branching ratio σ—the mean number of further agents each freshly-recruited agent goes on to recruit, so $\sigma < 1$ means cascades are finite and die out (App. C)—nearly flat (0.06–0.09), with its (broad) minimum near the operating point we use, and maintains near-ideal matching throughout (slope 0.80–1.03), with no per-condition tuning. A full Sobol sweep over all seven dimensionless controls confirms that the swarm's own parameters—the threshold ϕ and the Lévy exponent α—sit on broad plateaus (near-ideal slope across their swept ranges), while the gradient match depends mainly on how clustered the resource field is (App. F). We are careful about the sense of "critical." The single-seed recruitment cascade has the classic window between the percolation onset $z_c^{\mathrm{RGG}} \approx 4.51$ and the blocking bound $2/\phi$ (Fig. 5a); but because recruited agents aggregate, the swarm's *bulk* mean degree runs *above* that window, so the criticality is *local* (clusters near-critical, global branching subcritical), as the companion paper develops (Wilkerson, 2026). What is self-organized here is the *regime*—a density-invariant subcritical operating point and robust near-ideal matching, reached without tuning—not a bulk mean degree sitting inside the cascade window. "Not tuned" is a statement about fitting to the task, not about being parameter-free: the model's fixed constants—the Lévy exponent α and the activation threshold ϕ—set *where* in the regime the swarm operates, not *which* behaviours appear, and are held at a-priori values rather than selected to maximise harvest; indeed neither is at its harvest-optimal value, so the result is not an artefact of parameter search. This matters for the learner comparison (§6): the swarm meets the trained learner with parameters that were never fitted to the foraging task.

5 How it reaches the optimum: learner-free, reward-free, and not by learning

The swarm reaches the optimum. This section establishes *how it does not*: without a learner, without a reward signal, and without learning over time. These absences are the content of "learner-free," and we show each directly rather than asserting it.

No learner. There are no adjustable parameters and no training phase. Ordinary RL (Sutton & Barto, 2018) keeps a value function or a policy network and updates it over many episodes; the trained parameters

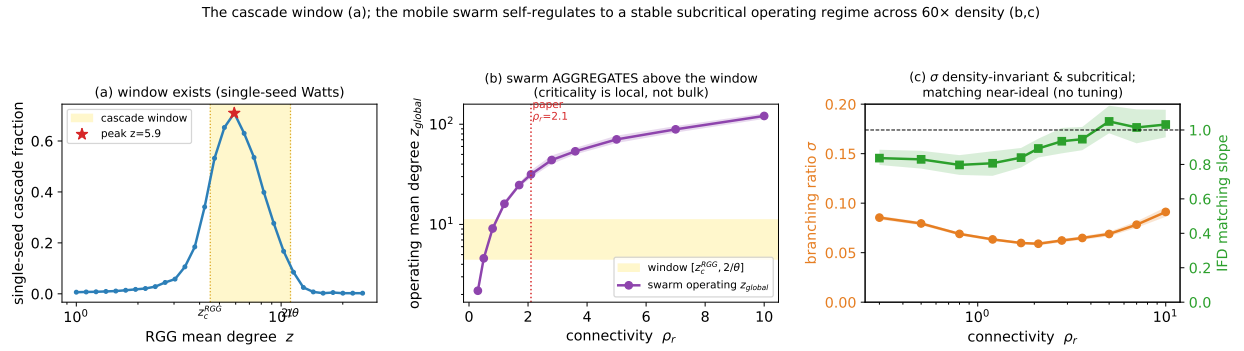


Figure 5: **The operating regime is self-organized, not tuned. (a)** The single-seed recruitment cascade has an inverted-U window between $z_c^{\mathrm{RGG}} \approx 4.51$ and the blocking bound $2/\phi$ (reproduced here). **(b)** Swept over a 60× connectivity range, the mobile swarm holds a subcritical branching ratio σ nearly flat (0.06–0.09), with its broad minimum near the operating point used throughout ($\rho_r = 2.1$); its bulk mean degree runs *above* the single-seed window (aggregation), so the criticality is local, not bulk. **(c)** Near-ideal IFD matching (slope 0.80–1.03) holds across the whole range with no per-condition tuning.

are the product. Here the instantaneous configuration $\mathbf{n}(t)$—carried by the agents' positions and their interaction network, stored nowhere else—*is* the allocation, taken in real time, with nothing trained or stored.

No reward signal: recruitment is reward-free by construction. The decisive evidence is the rule itself: per-capita intake f_i is never computed or read by any agent—a seed fires on contact with resource, and recruitment responds to the active *fraction* of neighbours (§3). There is simply no reward in the loop, so no reward can be followed; f_i is a quantity *we* compute to describe the outcome, not a signal the dynamics use. Measurements of the flow are consistent with this but, as we note below and in App. C, cannot by themselves establish it. Illustratively (Fig. 6), logging per-patch recruitment inflow over 64 replicas, inflow correlates *positively* with local occupancy n_i (Pearson $r = +0.11$)—the crowd signal the cascade rule uses—weakly *negatively* with per-capita payoff f_i ($r = -0.11$), and shows no monotone trend with richness Q_i (non-monotone, linear $r \approx 0$). The negative payoff relation is mechanical: at a patch $n_i f_i = Q_i/N$, so a patch recruitment has crowded necessarily has low per-capita intake. These correlations are small and illustrative, not an independent proof of reward-blindness—under equal sharing payoff-rank and distance-from-equilibrium coincide (App. C), so no correlational test can separate "reward-free" from "reward-following." The proof is the construction. The upshot is that the optimum is reached by a mechanism *blind to reward*: the agents pile onto richer patches only because richer patches, having more resource sites, are discovered and re-seeded more often, not because anyone measures that they pay better—the strongest sense of "without a learner," with no reward in the loop to learn from.

What the coarse dynamics are (and are not). Since the flow carries no reward, is it nonetheless a reward-following law in disguise? No. Fitting the denoised mean flow of the allocation over the transient to competing mean-field laws, by leave-replicas-out cross-validation (App. C), a reward-proportional *replicator* update is *not* the best description (held-out $R^2 = 0.26$); the flow is a *nonlinear relaxation* toward the ideal-free fixed point (held-out R^2 up to 0.64 for saturating forms). The coarse dynamics are thus a relaxation of the allocation to the optimum, not a reward-gradient ascent. We report this as a characterisation of the flow, not as an RL identification: the relaxation reaches the optimum; it is not a learning update.

It reaches the optimum, it does not learn. Reaching the optimum is not epistemic learning. The primary reason is architectural: there is no stored or accumulating state—no value estimate, no counts, no memory—so there is nothing in which experience could accumulate. Behaviourally this shows as two things (Fig. 7): under fixed conditions the allocation's informativeness about the resource field rises once and *plateaus* by ∼118 steps rather than improving with more experience; and when the resources are relocated

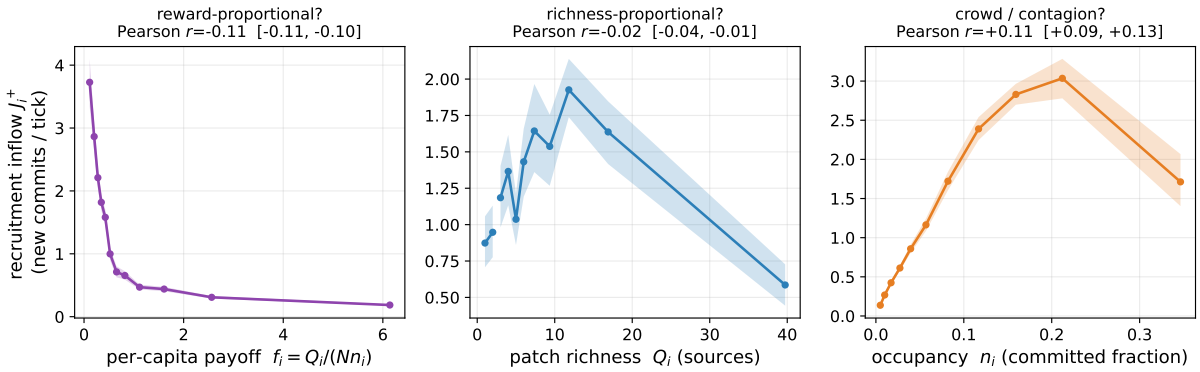


Figure 6: **Illustrative: recruitment tracks crowd, not payoff** (reward-freeness itself is established by the rule's construction, §3, not by this figure). Per-patch recruitment inflow J_i^+ (new commits per step) vs. per-capita payoff f_i, richness Q_i, and occupancy n_i (quantile-binned mean, bootstrap 95% CI over 64 replicas). Inflow correlates positively with occupancy ($r = +0.11$)—the crowd signal the cascade rule uses—weakly negatively with payoff ($r = -0.11$), and shows no monotone trend with richness (non-monotone, linear $r \approx 0$). The correlations are small and, under equal sharing, non-identifying (App. C); they illustrate rather than prove the reward-free construction.

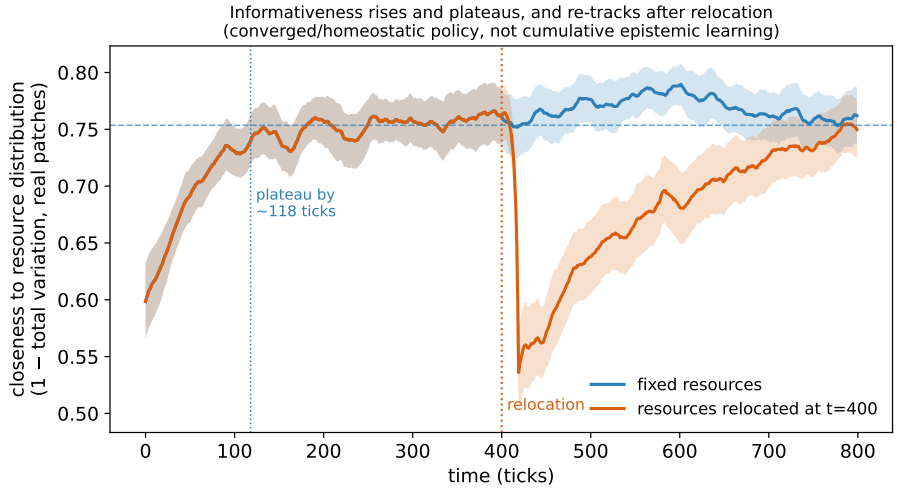


Figure 7: **The swarm reaches the optimum but does not learn it.** Allocation informativeness (closeness = 1− total-variation distance between committed occupancy and the resource distribution; replicas with bootstrap 95% CI) vs. time. It rises once and *plateaus* by ∼118 steps; after the resources are relocated (t=400, dashed line) it dips and *re-tracks* to the same plateau, not a higher one—no cumulative improvement. This is a converged policy, not a learner.

mid-run, the swarm re-settles to the *same* quality it held before, with no cumulative gain (relocation at t=400). (The informativeness metric is bounded, so a plateau alone is only suggestive; the load-bearing evidence is the absent accumulating state plus the re-tracking to the same, not a higher, level.) The swarm reaches and *holds* the optimum, re-finding it when the world changes—a converged, homeostatic policy, not an agent accumulating knowledge.

6 Comparison to a trained learner: who reaches the optimum

How does a learner-free mechanism compare to an actual learner? We compare in *two* settings. First, an *abstract* (disembodied) version of the congestion game—direct patch choice with a per-step payoff oracle and no spatial search—with independent softmax policy-gradient and Q-learning agents (App. D). Second, the *same embodied environment* the swarm acts in: one spatial world with identical local sensing, movement, and action set {explore, approach, stay}, where the *only* difference between arms is fixed (recruitment) versus learned control (App. E).

Disembodied, the learner wins on precision. On the abstract game the swarm reaches the right allocation *shape* at zero training (matching slope 0.95—consistent with the 0.94 in the spatial field, §4; 83% of the way to Nash by 1− total-variation distance; Fig. 8a) but is *not* near-optimal on exploitability (Nash-gap 0.58, roughly 50× a converged Q-learner's ≈ 0.011 and orders of magnitude above a policy-gradient learner's $\approx 5 \times 10^{-4}$; Fig. 8b). So a trained learner, handed a per-step payoff oracle and direct patch choice, buys an exploitability precision the swarm lacks. That advantage, though, is a feature of the disembodied game; the next experiment shows it does *not* carry over to the embodied, sensing-limited setting, where a trained policy reaches a *worse* allocation than the reward-free swarm.

Embodied and sensing-limited, the reward-free swarm is competitive on harvest and the most gradient-faithful in allocation. The decisive comparison is the *embodied* one, under primitives matched so neither arm can act on what it cannot sense. Both the swarm and the learners move through the same local sensing radius S: the *approach* action steers an agent only toward a resource it senses within S (or, for a recruited agent, along its neighbours' heading), never toward resource beyond its own range.[1] We score on two resource fields spanning the regimes a forager meets—a *scarce* field (sparse, hard-to-find resource) and a *clustered, resource-rich* field (patchy and heavy-tailed, the ecologically typical case)—by the fraction of a *validated oracle ceiling* (maximum intake under free placement given the current resource state, a strict upper bound no run approaches) and, separately, on two allocation measures: the ideal-free allocation slope (how faithfully committed foragers track richness, §4) and the exploitability (the occupancy-weighted coefficient of variation of per-capita intake across occupied patches; 0 at the equal-intake Nash point). We train three learner families: independent PPO (IPPO); a *communication-capable* graph policy running L rounds of learned message passing over the same proximity graph the cascade uses ($L \in \{1, 3, 5\}$); and, as the steelman, a centralized-training–decentralized-execution learner (MAPPO: the same sensing-limited actor as IPPO but trained against a centralized critic that sees privileged global state—the strongest learning signal a fair, sensing-limited actor can be given). Each is given a rich 14-dimensional local observation and a dense shaped reward, and trained to convergence (150,000 environment steps; the binding comparator, MLP-IPPO, is plateaued over the last half of training, Fig. 10); evaluation is always the true intake fraction (App. E). Over 16 seeds per learner (Fig. 9):

Harvest. In the clustered, resource-rich field the reward-free swarm *matches* the best trained learner outright—0.80 of the ceiling at zero training versus MLP-IPPO's 0.80. The 0.002 difference is not merely non-significant ($p = 0.55$, Welch's t over 16 seeds); the two are *statistically equivalent* by a two-one-sided-test: the 90% CI of the difference, $[-0.004, +0.008]$, lies entirely within a ± 0.02 equivalence margin ($p < 0.001$), so this is positive evidence of parity, not absence of evidence (random 0.32). We set the ± 0.02 tolerance a priori at $\approx 2\times$ the largest per-seed 95% CI ($\lesssim 0.01$; App. E) and 2.5% of the oracle ceiling; because the difference's own CI spans only $[-0.004, +0.008]$, the equivalence also holds under a margin half that size, so it is not an artefact of the chosen tolerance. It does so against the whole learner field: the centralized-critic steelman reaches only 0.76 (swarm ahead by 0.045, $p < 0.001$, Welch's t over 16 seeds) and the communicating graph policy 0.75. In the scarce field the trained learner collects *more* total intake (0.33 versus the swarm's 0.22, $p < 0.001$; random 0.14): where resource is hard to find, reading reward and training a policy buys raw harvest the reward-free mechanism does not—though the swarm still clears random by 59% (0.224 vs 0.141).

[1] An earlier version of this comparison let *approach* target the nearest live resource at *any* distance—a clairvoyant go-to-food primitive available to both arms. Because every agent could then head straight to resource it could not sense, the recruitment cascade's role—carrying a discovery through the network to agents that cannot see it—was short-circuited, and the comparison did not isolate what recruitment contributes. Restricting approach to sensed resource is the fair test, and the one reported here; it supersedes the earlier embodied numbers.

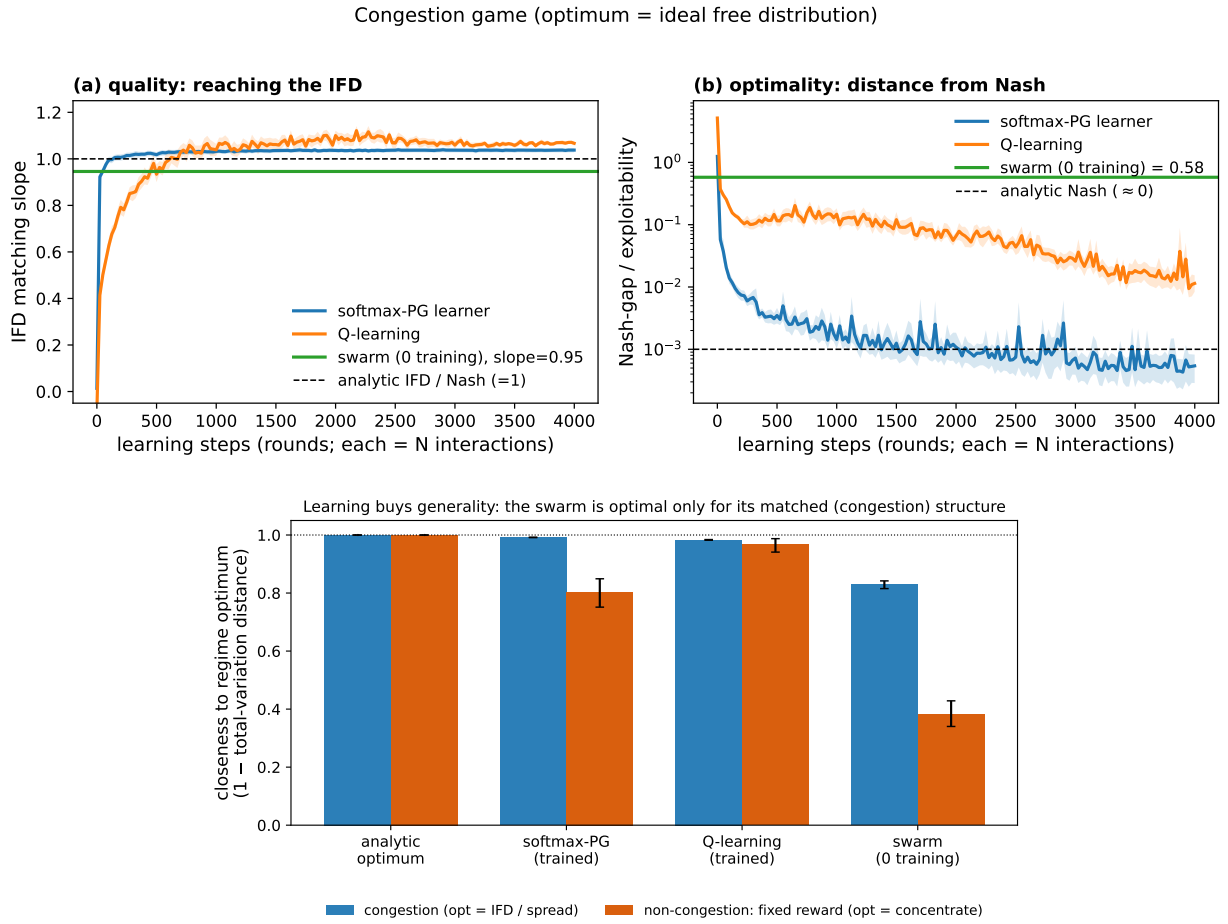


Figure 8: **Learner-free swarm vs. trained multi-agent RL, same optimum (abstract congestion game). (a)** matching slope and **(b)** Nash-gap (exploitability) vs. learning steps for independent softmax policy-gradient and Q-learning (bootstrap 95% CI over 64 replicas), with the zero-training swarm and the analytic optimum as references: the swarm reaches the right allocation *shape* at zero training (a), but its Nash-gap (0.58) stays well above the trained learners' (b)—it is not near-optimal on exploitability. The learners play an easier, disembodied problem (no spatial search); units differ (learner steps vs. swarm physical steps), as marked. **(c)** Closeness to the regime optimum in the *congestion* regime (optimum = ideal free distribution) and a *non-congestion* regime (fixed per-patch reward, optimum = richest patch). The swarm is near-optimal only in the congestion regime its mechanism matches; the learners adapt to both. Learning buys generality.

The swarm is thus competitive everywhere, statistically ties the best learner in the clustered regime, and is not out-harvested by the centralized-critic steelman there; we do *not* claim it out-harvests a trained learner universally.

Allocation: a bias–variance split. The allocation result is the one on-thesis, and it has two sides that must be reported together. On the *shape* of the allocation—how faithfully committed foragers track patch richness, the ideal-free matching slope—the reward-free swarm is the most faithful of the arms tested in both fields: its slope is 0.99 (scarce) and 1.02 (clustered), essentially the ideal 1, while every trained learner systematically *under*-concentrates (IPPO 0.76 and 0.91; the centralized-critic MAPPO 0.69 and 0.83; the communicating graph policy 0.77 and 0.86; Fig. 9b). The emergent allocation is thus *unbiased* with respect to the resource gradient; the trained policies are biased toward spreading more evenly than richness warrants. On the *precision* of the allocation—how tightly per-capita intake is equalized across occupied patches, the

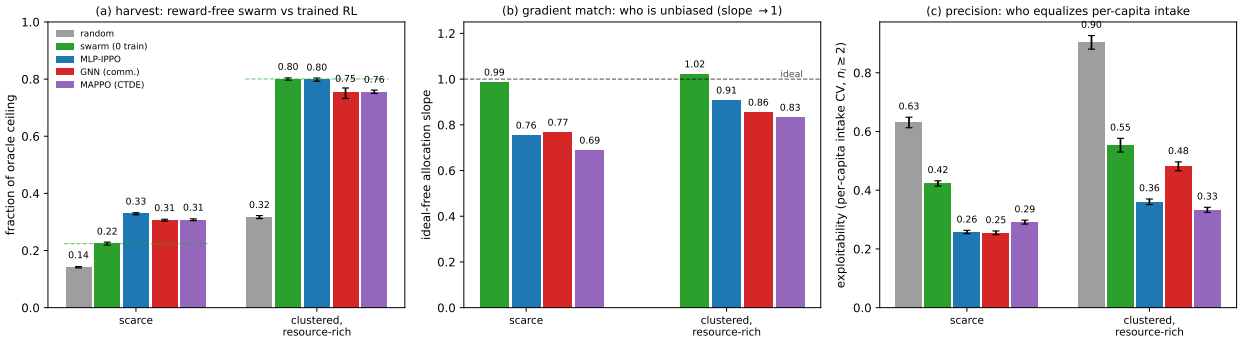


Figure 9: **Embodied comparison under matched, sensing-limited primitives: the reward-free swarm is competitive on harvest and the most gradient-faithful in allocation.** Same spatial environment; *approach* steers only toward resource sensed within radius S, identical for every arm; 16 seeds per learner, trained to convergence (150,000 env-steps); the communicating learner is a graph policy with a single round (L=1) of learned message passing (L=3, 5 do not help), and MAPPO is a centralized-critic (CTDE) learner. **(a)** Harvest, fraction of a validated oracle ceiling. Scarce field: swarm 0.22 (zero training), MLP-IPPO 0.33, GNN 0.31, MAPPO 0.31, random 0.14. Clustered, resource-rich field: swarm 0.80 is statistically equivalent to MLP-IPPO 0.80 (TOST within ± 0.02, $p < 0.001$) and beats MAPPO 0.76, GNN 0.75, random 0.32. The swarm is competitive everywhere and ties the best learner in the clustered regime; the learner out-harvests it only where resource is scarce. **(b)** Ideal-free allocation slope (dashed = ideal 1): the reward-free swarm matches the resource gradient most faithfully in *both* regimes (slope 0.99, 1.02)—it is unbiased w.r.t. the gradient, while every trained learner under-concentrates (IPPO 0.76, 0.91; GNN 0.77, 0.86; MAPPO 0.69, 0.83). **(c)** Exploitability (occupancy-weighted CV of per-capita intake over patches with $n_i \geq 2$; lower = tighter): here the ordering reverses—the trained learner equalizes per-capita intake more tightly than the swarm, which is nonetheless well below random. Panels (b) and (c) are the bias–variance split: emergent allocation is gradient-unbiased but noisier; learning buys precision, not a better gradient match.

exploitability—the ranking reverses: the trained learner is tighter (MLP-IPPO exploitability 0.26 scarce, 0.36 clustered) than the reward-free swarm (0.42 and 0.55), though the swarm still improves on random (0.63 and 0.90; Fig. 9c, per-capita CV over patches with $n_i \geq 2$ foragers; the ordering is stable across thresholds and under a Gini measure, App. E). The two measures do not contradict: an allocation can sit right on the richness gradient on average (slope $\approx$1) yet scatter around it from patch to patch, and the recruitment cascade's heavy-tailed occupancy produces exactly that scatter, whereas a trained policy trades a little gradient bias for lower per-capita variance. The honest reading is therefore a bias–variance split: *the reward-free mechanism reaches the correct allocation shape—the ideal-free optimum of the stated problem—at zero training and with no reward, more faithfully than any learner tested (whose reward shaping and entropy bonus, both spreading pressures, are held fixed rather than ablated; App. E); learning buys precision, not a better match to the gradient.*

Neither communication nor a centralized critic rescues the learner. Two obvious objections are that a feedforward learner, acting on its own observation, cannot represent the within-step coordination the cascade gets by propagating a discovery across the proximity graph, and that independent learners lack the joint credit assignment a centralized critic provides. We addressed both. Learned message passing did not help: the communicating graph policy underperforms even feedforward IPPO (clustered 0.75 vs 0.80; scarce 0.31 vs 0.33), and adding rounds does not improve it (L=3, 5 give 0.70/0.71 clustered, 0.29/0.26 scarce; Fig. 9a). The centralized-critic steelman did not help either: MAPPO, with privileged global state in its critic at training time, reaches 0.76 (clustered) and 0.31 (scarce)—at or below feedforward IPPO, and below the reward-free swarm in the clustered field. We read these cautiously—they show that *learned* message passing and a centralized critic, at this training budget, do not reach the reward-free cascade's allocation, not that no architecture ever could—but together they remove the obvious "the learner merely lacked communication

To reach the optimum	Learner-free swarm (this work)	Trained multi-agent RL
Reward signal	none (reward-free, Fig. 6)	per-agent scalar payoff, every step
Stored parameters	none	a policy (and critic) network
Training before use	none; acts at its operating point	150,000 env-steps with a dense shaped reward
Communication	local recruitment only (within range)	none (IPPO); learned message passing (GNN); centralized critic (MAPPO)
Movement to coordinate	none beyond exploration (recruits reuse the exploring step)	learned from local observations
Parameters tuned to the task	none; α, ϕ fixed a priori and *not* harvest-optimal	policy fit to a dense reward
Precision, *abstract* game	right shape (slope 0.95), but exploitable (Nash-gap 0.58)	near-exact (Nash-gap ≈0)
Harvest, *embodied* clustered field	0.80 of ceiling at *zero* training (Fig. 9)	0.80 (IPPO): statistically equivalent (TOST, ±0.02); 0.76 (MAPPO)
Harvest, *embodied* scarce field	0.22 at zero training	0.33 (IPPO): learner collects more
Allocation shape (gradient match), *embodied*	slope ≈1 in both regimes (most faithful, unbiased)	slope 0.69–0.91 (underconcentrates)
Allocation precision (per-capita CV), *embodied*	0.42/0.55: below random, above the learner	0.26/0.36 (IPPO): tighter per-capita intake
Transfer across reward structures	none (congestion/IFD only)	adapts

Table 2: **The two pay for different things.** The learner-free swarm reaches the allocation shape with no reward signal, no parameters, and no training—and its fixed parameters are not even tuned for harvest; the trained learner spends a reward signal, parameters, and training to buy precision (disembodied) and transfer. Under matched, sensing-limited embodiment the swarm's harvest is statistically equivalent to the best trained learner's in the clustered, resource-rich regime and matches the resource gradient most faithfully (slope ≈1, unbiased) in both regimes, while the learner collects more raw intake only where resource is scarce and buys lower per-capita variance in both—a bias–variance split in allocation (Fig. 9).

/ joint credit assignment" objections. The honest summary of the embodied contest is therefore: matched for sensing, the reward-free swarm is competitive with trained PPO on harvest (statistically tying the best learner in resource-rich fields, behind only where resource is scarce) and matches the resource gradient most faithfully of the arms tested in both fields (slope ≈1), at zero training, no reward, and no parameters—while a trained learner buys more raw harvest only in the scarce regime and lower per-capita variance in both, and neither learned communication nor a centralized critic changes the picture; nor does scaling the learner's capacity (4× width) or learning rate, which moves the clustered-field ceiling by <0.01 (App. E), so the equivalence is not against an under-powered baseline. This is one pair of resource fields and three learner families, which *bounds* rather than settles the general comparison.

What a learner still buys: transfer. One advantage of learning is unambiguous. In the abstract game's non-congestion regime (fixed per-patch reward, optimum = concentrate on the richest patch) the learners adapt (80–97% of optimum) while the reward-blind swarm keeps spreading and is sub-optimal (38%; Fig. 8c). A learner, by reading reward, transfers across reward structures; the fixed, reward-blind swarm cannot. (The embodied environment does not provide a clean non-congestion adaptation test—intake there accrues per agent-on-source—so we rest the transfer point on the abstract game.)

7 Discussion

Where the reinforcement learning is. The identification is at the level of the problem and the allocation reached, not the micro-rule (established in §5): the swarm attains the *shape* of the optimum's allocation—though not its exact equilibrium, remaining measurably exploitable (§6)—while evaluating no reward and

without learning. So "learner-free reinforcement learning" is a careful statement, not a relabelling: the problem is reinforcement learning's, the outcome is (approximately) its optimum, and the learner is absent.

Why this is more than aggregation. A sceptic may ask whether "reaching the ideal free distribution" is just agents piling onto richer patches. Three things make it a genuine optimum rather than a trivial pile-up. First, the allocation it reaches has the *shape* of the optimum (foragers in proportion to richness, approaching equal per-capita intake), not an arbitrary aggregation—and the ablation shows it is non-trivial to reach (without recruitment the swarm does *not* distribute proportionally, slope 0.08). Second, the explore–exploit trade-off is actively and continually balanced by the cascade cycle (§3), not set once. Third, the optimum is an attractor: the swarm re-finds it when the resources move (Fig. 7).

Limitations. (i) The optimum reached is the ideal free distribution of a resource-sharing (congestion) problem; the swarm's mechanism is matched to that structure and does not transfer to others (§6). That structure, however—sparse, patchily clustered resources (heavy-tailed in size and spacing) shared among foragers—is the common case in natural foraging environments, not a special one, so the domain the mechanism is matched to is broad and ecologically typical rather than contrived. (ii) Two forces set the allocation, and we separate them. The density-dependent dispersal ("blocking") regulates the explore–exploit *balance* and the critical operating point—shown here by the self-tuned subcritical regime across a 60× density range (Fig. 5), and mapped in full across connectivity and threshold in the companion paper (Wilkerson, 2026). The allocation *shape* (slope-one matching), by contrast, is set by richness-proportional discovery and recruitment, not by the crowding cutoff—the cutoff-removal ablation in §4 (0.94 → 0.92) makes this separation explicit, a consistency check alongside Part I's swept characterisation. (iii) The trained-learner comparison includes both an abstract game and an embodied, partially-observed contest under sensing-limited primitives matched across arms (§6); in the embodied setting the reward-free swarm's harvest is statistically equivalent to the best trained learner's in the clustered regime and matches the resource gradient most faithfully of the arms tested in both regimes, while the learner out-harvests it only where resource is scarce and buys lower per-capita variance (Fig. 9), so the harvest claim is regime-dependent rather than universal and the allocation picture is a bias–variance split rather than a clean win for either. It is one pair of resource fields and three learner families (independent PPO, a learned-message-passing graph policy, and a centralised-training–decentralised-execution learner with a privileged critic, MAPPO), so it *bounds* rather than settles the general swarm-vs-RL question: a fully tuned state-of-the-art MARL sweep is beyond our scope, but the principled steelman—a centralised critic, the strongest signal a fair sensing-limited actor can be trained against—is included and does not overturn the comparison. (iv) We give no closed-form regret or convergence-rate bound.

Computational cost. The comparison also has an asymptotic cost axis, orthogonal to the harvest outcome. At deployment all methods are linear in the number of agents N per step: the swarm's work is $\Theta(N)$ (each agent senses and recruits within a bounded neighbourhood—expected degree $\rho_r = N(r/L)^2$ is $O(1)$—and with spatial binning the neighbour graph is built in $\Theta(N)$; the recruitment cascade is subcritical, branching ratio $\sigma \approx 0.06$–0.09, so its total propagated work per step is $\Theta\big(N/(1-\sigma)\big) = \Theta(N)$), the same order as one environment step and as a learner's forward pass ($\Theta(Nh^2)$ for hidden width h, or $\Theta(L\,Nh^2)$ with L message-passing rounds). The difference is *offline*: the swarm trains on zero samples and stores no parameters, while each learner consumes 150,000 environment steps of training—$\Theta(TNh^2)$ work for T steps—and a stored policy (and critic). Computing the ideal-free optimum *centrally* is cheap in the abstract ($\Theta(M)$ water-filling over M sources), but requires global knowledge of every patch's richness and occupancy; the swarm reaches the same allocation *shape* from local information alone, with no central solve and no training. Cost is therefore not where the swarm loses: it is strictly cheaper offline and no worse than linear online. What a learner's training budget buys is the harvest edge in the scarce regime, the tighter per-capita precision, and transfer—not a cheaper deployment.

Outlook. Under matched embodiment the picture is regime-dependent on harvest (the swarm ties the best trained learner in resource-rich fields and trails it only where resource is scarce) and a bias–variance split on allocation (the swarm matches the resource gradient most faithfully everywhere, the learner equalizes per-capita intake more tightly; Fig. 9). The natural next steps are to characterise *where* the learner's harvest

edge appears—how it moves with resource scarcity and search difficulty, nonstationarity, agent count, and learner architecture—and to give a convergence-rate / regret characterisation of the physical flow against standard bandit and MARL algorithms, including whether a hybrid that warm-starts a learner from the swarm's zero-shot allocation shortens its budget.

Why it matters. As an engineering primitive, the mechanism delivers a close approximation of the explore–exploit optimum—a load-balanced, near-ideal-free allocation—with no model, no reward signal, no training, and no fitted parameters (its fixed α, ϕ are not even tuned for harvest), deployable on minimal agents. Matched for embodiment it matches the ideal-free allocation gradient *more faithfully* than a trained learner and ties the learner's harvest in resource-rich fields, at zero training; a well-trained learner—given a reward signal and a budget—collects more raw intake only where resource is scarce, and buys a tighter per-capita intake (§6). It is a case where the optimisation *is* the physics of a self-organizing collective, not an algorithm run on top of it.

Deployment without training. The absence of a learner has a direct operational consequence: the swarm can be released into a new environment with no training phase, no reward engineering, and no prior map of where the resource lies. Consider autonomous drones dispatched to locate and sample sparse, clustered atmospheric turbulence. The swarm collects no training data and requires no model of the turbulence field; it organizes its spatial allocation online from whatever it encounters. The self-organization is of the *allocation*—where the agents concentrate given what they find—not of the control parameters themselves.

This does not make the system parameter-free, but the parameters fixed before launch fall into benign, transferable classes. The first is set by hardware and geometry and is measured rather than tuned: the fleet size N, the survey scale L (area $A = L^2$), the sensing radius S, and the communication radius r follow directly from the platform and the mission area. Because the dynamics depend only on dimensionless combinations of these, the single design check is that the connectivity density $\rho_r = N(r/L)^2$—the mean number of neighbours within communication range—lands in the operating range identified here and in the companion paper (Wilkerson, 2026); the sensing radius enters only through the ratio S/L, the fraction of the arena a single drone perceives. The raw scales themselves do not matter—only these ratios, which is what lets the settings transfer across missions. The second is a single dimensionless control, the recruitment threshold ϕ (which sets the cascade branching ratio σ), placed in the subcritical-but-active regime (here $\sigma \approx 0.06$–0.09); because the useful range is a broad plateau rather than a knife-edge and is dimensionless, it transfers across settings and can be fixed once from general principles. The search movement is likewise a transferable default: an inverse-square Lévy search (stability index $\alpha = 1$, giving a step-length tail exponent $\mu = 1 + \alpha \approx 2$) is close to optimal for sparse targets across a wide range of conditions. Its optimum does shift with the clustering of the resource field (Wilkerson, 2026), but sits on a broad plateau, so it too is set once rather than retuned per deployment and is not adapted online. One domain-specific modeling step remains: the allocation mechanism relies on local negative feedback—in foraging, resource depletion and congestion. For a non-consumable resource such as turbulence, this role is played by coverage or information saturation, the declining marginal value of an additional drone at an already well-sampled location. In short, the swarm removes the need for training and for prior knowledge of the resource field; it does not remove the need to set a handful of transferable, interpretable controls.

8 Conclusion

A foraging swarm faces the explore–exploit problem that defines reinforcement learning. We have shown that a minimal self-organizing swarm *approaches its optimum*—balancing exploration and exploitation through its cascade cycle and converging to the *shape* of the optimal allocation of foragers over patches (the ideal free distribution)—with no learner, no trained parameters, and no reward signal in its dynamics, and that it does so by *reaching* this allocation rather than by *learning* it. Given a reward signal and a training budget, a learner reaches the abstract optimum more precisely, collects more raw intake where resource is scarce, equalizes per-capita intake more tightly, and transfers to other reward structures; the reward-free swarm matches the ideal-free allocation gradient at least as faithfully—more faithfully than any learner, under matched embodiment—at none of that cost, and ties the best learner's harvest where resource is

clustered: the price and the point of having no learner. The reinforcement learning here is the problem and the (approximate) optimum reached, emergent from physics.

Competing interests

The author has filed a provisional patent application related to the recruitment-switching mechanism described in this work.

Data and code availability

Complete simulation code, all configuration files, precomputed sweep and comparison data, figure-regenerating scripts, and an interactive browser-based demo of the mechanism are released publicly with the paper (an anonymized copy accompanies the submission for review).

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A Complete model specification

This appendix specifies the model in full, so the paper is reproducible without the companion. All constants are in Table 1; the embodied-comparison operating points and the learner setup are in App. E. Code and configuration are released (Data and code availability).

Arena and agents. N=200 agents live on a square torus $[0, L)^2$, L=800, with positions initialised uniformly at random. Time is discrete; each step is one movement plus one recruitment pass. The embodied comparison runs a horizon T=600 steps.

Resource field. There are C=10 resource clumps with centres drawn uniformly on the torus. Clump richness is heavy-tailed: M=80 point sources are assigned to clumps by a Pareto-like weighting—weights $w_c \propto U_c^{-1/(\beta-1)}$ (with $U_c \sim \mathrm{Unif}(0, 1]$, exponent β=2.2) normalised to a distribution over clumps, then each source assigned by an independent draw—so a clump's richness Q_c (its number of sources) follows a Pareto-like law over a ∼40-fold range. Each source is placed at its clump centre plus an isotropic Gaussian offset of standard deviation clump_radius $= \rho_{\mathrm{patch}} L$ (the patchiness ρ_{patch} is an operating-point parameter; App. E), wrapped onto the torus. Every source holds capacity κ=40.

Two interaction ranges. The model uses two radii. (i) The *proximity graph* joins two agents within $r = L\sqrt{\rho_r/N} \approx 82$ (ρ_r=2.1); this graph carries recruitment, and an agent's degree k is its number of neighbours on it. (ii) The *resource-sensing radius* $S = L\sqrt{\rho_s/M}$ sets how close an agent must be to detect a source (ρ_s is the resource coverage, an operating-point parameter, as in the companion paper; App. E; $S \approx 12.6$ scarce, 25.3 clustered). An agent is *on food* when a live source ($\kappa_s > 0$) lies within S.

Movement. Each step an agent takes one of three moves. **Explore**: a displacement of heavy-tailed length in a uniformly random direction—the length is drawn from a one-sided α-stable law (Mantegna's algorithm) with exponent α=1.0 and scale 2.0. **Approach**: a step of the same Lévy length toward the agent's current *target* (a sensed source, or, if recruited, the recruitment heading below). **Stay** (local search): an isotropic Gaussian jitter of standard deviation 3. Positions wrap on the torus. The reward-free swarm's controller chooses explore/approach/stay by the rule below; a learner replaces that rule with a trained policy over the same three moves (App. E).

Recruitment cascade (linear threshold, multi-hop). Recruitment is a synchronous linear-threshold cascade on the proximity graph, run to a fixed point each step. The *seeds* are the on-food agents. An inactive agent i *activates* when the active fraction of its neighbours reaches the threshold,

$$\frac{a_i}{k_i} \;\geq\; \phi, \qquad \phi = 0.18,$$

where a_i is the number of active neighbours and k_i the degree; rounds repeat until no new agent activates. There is no per-hop signal decay (equivalently a relay factor of 1 at every hop): an active neighbour contributes equally regardless of how far down the cascade it was recruited. A recruited agent (active but not itself on food) takes the *approach* move along the centroid heading of its *parents*—its active neighbours that activated in an earlier round—so a discovery is carried outward through the network to agents that cannot sense it. Recruited agents reuse the exploring step length and change only direction, so no extra movement budget is spent to coordinate.

Density-dependent dispersal (blocking). Blocking is not a separate rule; it is a consequence of the threshold being *degree-normalized*. As a neighbourhood crowds, each agent gains neighbours faster than it gains *active* neighbours, so a_i/k_i falls back below ϕ and no further agents are recruited to the crowded region: the same rule that spreads a cascade stifles it once a patch is worked, and the swarm releases agents back to exploration.

Consumption, congestion, and depletion. An on-food agent draws from the single nearest live source it is within S of. The resource a source yields in a step is shared equally among the agents drawing on it: with n agents on a source of capacity κ_s, the source yields $\min(\kappa_s,\, c\,n)$ with per-agent consumption c=1, so each of the n agents receives $\min(\kappa_s, cn)/n$ and the source's capacity is decremented by the amount yielded. Per-capita intake therefore falls as a source crowds—the congestion that makes the optimum an allocation. When a source's capacity reaches 0 it *respawns* immediately: it is re-placed at a fresh Gaussian offset around its clump and its capacity is reset to κ, so the field is stationary in distribution while individual sources deplete and move.

Branching ratio σ. We summarise cascade spread by the branching ratio σ: the mean number of secondary activations triggered per active agent over a cascade (the expected number of agents a freshly-activated agent goes on to activate). $\sigma < 1$ is *subcritical*—cascades are finite and die out; $\sigma \approx 1$ is critical. Across the swept density range the mobile swarm holds $\sigma \approx 0.06$–0.09 (Fig. 5), i.e. globally subcritical.

Per-step update (summary). (1) sense sources within S and build the proximity graph within r; (2) run the linear-threshold cascade to its fixed point to label on-food/recruited agents and recruitment headings; (3) choose each agent's move (swarm rule, or learned policy); (4) apply moves with torus wrapping; (5) resolve shared consumption, decrement capacities, and respawn depleted sources. The configuration carried by the agents' positions and the proximity graph—stored nowhere else—*is* the swarm's allocation at that step.

Dimensionless form. Every raw scale above enters the dynamics only through dimensionless combinations, so the model is governed by seven dimensionless controls—the axes of the robustness sweep (App. F). Three are the swarm's *own* controls, held at task-independent defaults (ρ_r, ϕ, α); three are properties of the resource *field* $(\rho_s, \rho_{\text{patch}}, \beta)$; and the relay factor γ is fixed at 1 (no per-hop decay) throughout this paper.

control	meaning	operating value	swept range
$\rho_r = N(r/L)^2$	connectivity (mean degree)	2.1	$[1.0, 5.0]$
ϕ	activation threshold ($a/k \geq \phi$)	0.18	$[0.05, 0.35]$
α	Lévy exponent	1.0	$[0.6, 2.0]$
β	Pareto patch-richness exponent	2.2	$[1.5, 3.5]$
$\rho_s = M(S/L)^2$	resource coverage	0.02–0.08	$[0.005, 0.08]$
ρ_{patch}	resource patchiness (clump radius/L)	0.0375	$[0.02, 0.30]$
γ	relay decay (per hop)	1.0	$[0.1, 1.0]$

B The optimal allocation: ideal free distribution and habitat-game Nash

With resource Q_i shared among the Nn_i foragers on patch i, per-capita intake is $f_i = Q_i/(Nn_i)$. An allocation is stable against unilateral relocation when per-capita intake is equal across all occupied patches, $f_i = \bar{f}$ for all occupied i. With $f_i = Q_i/(Nn_i)$ this gives $n_i \propto Q_i$: the ideal free distribution (Fretwell & Lucas, 1969). Because each agent's payoff depends only on how many share its patch, the habitat-selection game is an anonymous congestion game, whose pure Nash equilibrium is exactly this equal-intake allocation (Rosenthal, 1973; Wardrop, 1952); player-specific congestion games need not share this structure (Milchtaich, 1996). Note that mean per-capita intake $\langle f \rangle = \sum_i n_i f_i = \sum_i Q_i/N$ is constant on the interior, so the optimum is an equal-intake *equilibrium* (a proportional-fair, log-utility allocation (Kelly et al., 1998)), not a total-intake maximum: the swarm load-balances, it does not increase total intake. We use the ideal free distribution purely to *characterise* the solution the swarm reaches (slope-one matching, $n_i \propto Q_i$); the swarm does not compute payoffs or solve the game (§5).

C What the coarse dynamics are: model competition and the role of blocking

Model competition. We ask which mean-field law best describes the measured flow of the allocation $\dot{n}_i$. Per-step $\dot{n}_i$ is dominated by discrete-event noise (all candidate laws sit at held-out $R^2 \approx 0.03$ at step resolution), so we fit the *denoised mean flow*: samples binned by (richness stratum × occupancy), each model fit to the per-bin mean $\dot{n}_i$ over the transient (steps 1–150), with model selection by leave-16-replicas-out cross-validation and AIC/BIC. Candidates: a reward-proportional *replicator* $\dot{n}_i = r_1\, n_i(f_i - \bar{f})$ (which on equal sharing is algebraically a *linear relaxation* to the fixed point, $r_1'(n_i^* - n_i)$, $n_i^* = Q_i/\sum_j Q_j$); a saturating *best-response*/logit $\dot{n}_i = r_2\,(\text{softmax}_\tau(f)_i - n_i)$; a payoff-blind *logistic/contagion* $\dot{n}_i = r_3\, n_i(c_0 Q_i - n_i)$; and a constant null. (Here r_1, r_2, r_3, c_0 and the logit inverse-temperature τ are fitted rate constants, distinct from the model parameters of App. A.) Held-out R^2: logit 0.64, contagion 0.47, replicator/linear-relaxation 0.26, null 0. Two honest conclusions follow. (i) The linear replicator/relaxation form is *rejected* as sufficient—the flow has saturating curvature. (ii) The best-fitting forms are nonlinear *relaxations* toward the $n_i \propto Q_i$ fixed point; because on equal sharing payoff-rank equals gap-rank, the logit fit does not indicate that the swarm computes payoffs (it cannot—§5). The flow is a nonlinear relaxation of the allocation to the optimum, consistent with a reward-free mechanism. Two caveats keep this honest. First, under equal sharing the replicator regressor $n_i(f_i - \bar{f})$ equals the signed distance to the fixed point, $(Q_{\text{tot}}/N)(n_i^* - n_i)$ (measured correlation 0.997), so payoff-rank and gap-rank coincide and *no* regression on this data can separate a reward-free from a reward-following rule—reward-freeness rests on the rule's construction (§3), not on these fits. Second, the payoff-blind logistic/contagion form, though competitive in bulk R^2, is itself rejected by a shape diagnostic: at small occupancy the measured flow grows with the gap rather than being throttled toward zero as $n_i(c_0 Q_i - n_i)$ would require. So the fits identify "nonlinear relaxation to the fixed point" but neither a specific reward-following law nor a specific payoff-blind one.

The role of density-dependent dispersal (blocking). We separate blocking's two possible roles: setting the explore–exploit *balance* versus setting the allocation *shape*. For the balance, blocking is causal: we show in-paper that the swarm self-tunes to a subcritical operating regime across a 60× density range with no per-condition tuning (§4, Fig. 5), and the companion paper maps the bounding cascade window in full with phase diagrams over connectivity and activation threshold (percolation onset below, degree-dependent blocking above) (Wilkerson, 2026). For the allocation shape, we run a single intermediate ablation here: removing the crowding cutoff (replacing the degree-normalized rule $a/k \geq \phi$ with an absolute-count rule, the maximal

no-cutoff limit) while leaving recruitment otherwise intact. The matching slope is essentially unchanged ($0.94 \rightarrow 0.92$, no over-aggregation), while the active fraction rises ($0.73 \rightarrow 0.85$)—so blocking was holding more of the swarm exploring (it regulates the balance) but was not setting the proportional allocation. That shape traces to richness-proportional discovery and recruitment (richer patches, with more resource sites, are seeded more often). This is a single-operating-point consistency check, not a sweep (the self-tuning across conditions is Fig. 5; the full parameter map is in (Wilkerson, 2026)), and the absolute-count toggle is an extreme limit—but together they support attributing the allocation shape to the recruitment–movement coupling and discovery, while the self-organization (blocking) does the explore–exploit balancing.

D Trained multi-agent RL baseline

On an abstract version of the congestion allocation game (K patches, N agents, per-capita payoff $f_i = Q_i/(Nn_i)$, richness Q_i matched to the swarm's), each agent runs an independent learner with local, own-payoff information only: a softmax policy-gradient (REINFORCE with a per-agent baseline) or an ϵ-greedy Q-learner. Quality is the matching slope and the distance to the ideal free distribution; the "regret" analogue is the *exploitability* / Nash-gap, since reward-regret is ill-defined here (mean per-capita intake is constant, App. B). The swarm, being reward-blind, is run at its physical operating point and its steady-state allocation compared on the same (Q_i, N, K). The non-congestion regime replaces the payoff with a fixed per-patch reward $f_i = Q_i$ (optimum = concentrate on $\arg\max_i Q_i$), where the swarm's fixed mechanism is sub-optimal while the learners adapt. Learner hyperparameters are lightly tuned so the baseline is neither crippled nor over-fit; the swarm has no tunable parameters, an asymmetry we state rather than hide.

E Embodied learner comparison

All arms share one spatial environment: a torus with $N = 200$ agents per replica world, a local sensing radius S, Pareto-clumped resources that deplete and respawn, and the action set {explore (a Lévy step), approach, stay (local search)}. **Matched, sensing-limited primitives.** The *approach* action steers an agent only toward a resource it senses *within* S (or, if recruited, along its active neighbours' heading); no agent can head toward resource beyond its own sensing range. This is identical for every arm. (An earlier version let *approach* target the nearest live resource at *any* range—a clairvoyant primitive, shared by both arms, that let every agent walk straight to unsensed resource and so short-circuited the recruitment cascade; those numbers are superseded here.) The *swarm* arm applies the fixed recruitment+blocking rule over this observation/action set, reward-free; each *learner* arm replaces the rule with a trained policy over the same action set. Only the controller differs—sensing, movement, and actions are identical—so the comparison isolates fixed versus learned control under matched embodiment and partial observability, and because both arms use the same movement primitives and step budget the fairness does not rely on the companion paper's movement-neutrality claim. The swarm's fixed parameters are moreover *not* tuned to this task: the Lévy exponent α=1.0 and the activation threshold ϕ=0.18 are held at their a-priori values across all conditions and are demonstrably *not* harvest-optimal (a higher ϕ and a heavier-tailed α each raise intake), so parameter selection does not favour the swarm.

Operating points. We evaluate on two resource fields: SCARCE (low resource density, more dispersed) and CLUSTERED (resource-rich, tightly patched—the ecologically typical heavy-tailed case). The two differ only in environment/sensing parameters (resource coverage ρ_s, clump radius, congestion strength); α and ϕ are identical in both.

Learners. We train two policies by independent PPO (clipped surrogate, generalised advantage estimation, entropy bonus, a shared-weight actor with a per-agent local critic). *MLP-IPPO* is a feedforward policy on the egocentric observation. The *communication-capable* policy (GNN) runs L rounds of learned message passing over the same proximity graph the cascade uses before acting ($L \in \{1, 3, 5\}$), so it can in principle propagate a discovery across the network within a step as the cascade does. Both see a rich 14-dimensional egocentric observation (on-resource flag; nearest sensed-resource proximity and direction; local and wide resource density; neighbour-foraging fraction and local degree; nearest on-food neighbour proximity and

direction; own short-memory intake and on-food fraction)—at least as informative as what the swarm effectively uses, so the learner is advantaged on information, not handicapped. Training uses a dense shaped reward (per-step per-capita intake plus a small resource-proximity term); *evaluation always uses the true intake fraction of the oracle ceiling* $\min(N, \sum \text{cap})$ per step, never the shaped reward. Hyperparameters (shared): learning rate 3×10^{-4}, hidden width 64, rollout 128, 4 epochs, clip 0.2, $\gamma = 0.99$, $\lambda = 0.95$, value coefficient 0.5, entropy 0.01, gradient-norm clip 0.5, Adam.

What is held fixed in the learners. The shaped reward's resource-proximity term and the entropy bonus (0.01) both push a policy to *spread*, and we hold both fixed rather than ablating them (the capacity/learning-rate sweep below likewise does not vary them). The learners' systematic under-concentration (slope 0.69–0.91) is therefore reported *under this standard reward shaping*, and some of it may reflect these spreading pressures rather than an intrinsic limit of trained RL. The swarm's own gradient-unbiasedness (slope ≈1) has no reward to shape and does not depend on this, so the comparison we draw is the one-sided one—the reward-free swarm is at least as gradient-faithful as these learners as configured—not that trained RL is intrinsically biased. Ablating the shaping and entropy is left to future work.

Efficient GPU implementation. The whole environment is a batched GPU (PyTorch) port of the NumPy environment; every independent unit—16 seeds × 2 operating points × learner family—is folded into a leading ensemble dimension with grouped, per-policy weights, so the full comparison runs as one high-utilisation GPU job rather than a sequential sweep. The port is validated against the original NumPy environment: the reward-free multi-hop swarm reaches the same oracle fraction within seed noise and the same ideal-free slope, and random actions match.

Protocol and result. Each learner is trained for 150,000 environment steps—10× the earlier budget, long enough that the binding comparator (MLP-IPPO) is plateaued over the last half of training (relative gain <0.3% over the final quarter in both fields; Fig. 10)—and scored on a trailing evaluation window; we run 16 seeds per policy (the seed is the unit of uncertainty, since the shared-weight policy couples replicas) and compare against multi-seed reward-free swarm and random-action references on the same fields. We evaluate a third learner family as a steelman: MAPPO, a centralized-training–decentralized-execution learner whose actor is the identical sensing-limited IPPO actor but trained against a centralized critic that sees privileged global state (so the execution-time comparison stays fair while the training signal is the strongest a fair actor can be given). Fractions of the oracle ceiling (mean over 16 seeds; per-seed 95% CIs $\lesssim 0.01$ except the GNN, $\lesssim 0.03$):

	random	swarm (0 train)	MLP-IPPO	MAPPO (CTDE)	GNN (L=1/3/5)
SCARCE	0.14	0.22 (slope 0.99)	0.33 (slope 0.76)	0.31 (slope 0.69)	0.31/0.29/0.26
CLUSTERED	0.32	0.80 (slope 1.02)	0.80 (slope 0.91)	0.76 (slope 0.83)	0.75/0.70/0.71

In the clustered field the reward-free swarm's harvest is statistically *equivalent* to the best trained learner's (0.80 vs MLP-IPPO's 0.80; a two-one-sided-test rejects a difference beyond ±0.02 at $p < 0.001$, 90% CI of the difference $[-0.004, +0.008]$, 16 seeds) and is not out-harvested by the MAPPO steelman (0.76); in the scarce field the learner harvests more (0.33 vs 0.22, $p < 0.001$, 16 seeds). The allocation result is a bias–variance split. On the allocation *shape* (ideal-free matching slope) the swarm is the most faithful arm in both fields (slope ≈1 vs the learners' 0.69–0.91): it is unbiased with respect to the resource gradient while the learners under-concentrate. On allocation *precision* (exploitability = occupancy-weighted CV of per-capita intake over patches with $n_i \geq 2$) the ranking reverses:

exploitability (per-capita CV, $n_i \geq 2$)	random	swarm	MLP-IPPO	GNN (L=1)
SCARCE	0.63	0.42	0.26	0.26
CLUSTERED	0.90	0.55	0.36	0.48

The trained learner equalizes per-capita intake more tightly than the swarm, though the swarm remains well below random; learning buys precision, not a better gradient match. (Robustness: the stored metric uses the occupancy floor $n_i \geq 0.5$, at which a CV of the ratio Q_i/n_i is inflated by marginally-occupied patches—the swarm's heavy-tailed occupancy then spuriously pushes its clustered-field CV above random.

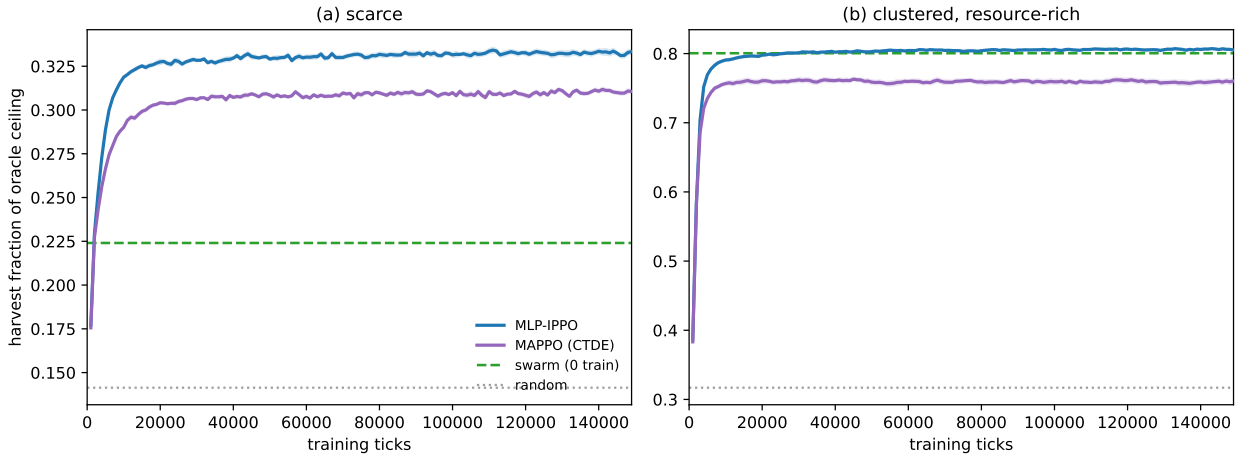


Figure 10: **The learners are trained to convergence (fairness check).** Harvest fraction of the oracle ceiling vs training ticks for the embodied learners (MLP-IPPO and the MAPPO centralized-critic steelman), with the learner-free swarm and the random baseline as horizontal references, in the **(a)** scarce and **(b)** clustered fields (16 seeds; bands are ±SEM over seeds). Both learners plateau well before the 150,000-step budget—MLP-IPPO's relative gain over the final quarter is <0.3% in both fields, slope $\approx 2 \times 10^{-5}$ per 1000 ticks over the last half—so the learner had every opportunity to converge before being compared to the zero-training swarm.

The ordering reported here, learner < swarm < random, is the one that is stable across occupancy thresholds, $n_i \geq 0.5/2/4$, and under a Gini measure of per-capita dispersion; we therefore report the robust $n_i \geq 2$ figure.) The communication-capable GNN does not beat the swarm and underperforms even the feedforward MLP-IPPO, adding message-passing rounds (L=3, 5) does not help, and the MAPPO centralized critic does not help either—so neither communication nor joint credit assignment overturns the comparison. Correctness checks: the learners rise well above the random baseline (they are clearly learning), and the same PPO implementation climbs the ideal-free matching slope on the easy abstract congestion game (App. D), confirming the optimiser is sound.

The baseline is not under-powered. To confirm that the learner's ceiling (≈0.80 clustered, ≈0.33 scarce) is not an artefact of an under-capacity or under-tuned network, we retrained the feedforward MLP-IPPO learner across hidden width (64 → 128 → 256, up to 4× capacity) and learning rate ($3 \times 10^{-4} \rightarrow 10^{-3}$), 8 seeds each. Capacity and tuning move the ceiling by less than 0.01: the clustered-field fraction stays 0.80–0.81 (width-256 reaches 0.808) and the scarce-field 0.329–0.332. The reward-free swarm's clustered-field equivalence (0.801) therefore holds against the strongest learner we trained—within the ±0.02 margin—not merely the default one, so the comparison is not against an under-powered baseline.

A note on earlier versions: a deliberately simple online REINFORCE controller once plateaued near random and was mistaken for evidence that an embodied learner cannot match the swarm; a later, properly-trained comparison instead *over*-credited the learner because it ran under the clairvoyant any-range approach primitive. The sensing-limited comparison here supersedes both—the swarm is competitive on harvest (statistically equivalent to the best learner where resource is clustered) and the most gradient-faithful of the arms tested in both fields, the learner is more precise on per-capita intake, and neither learned communication nor a centralized critic changes that.

F Robustness of the allocation across the dimensionless controls

To check that the near-ideal allocation is not an artefact of the single operating point, we ran a low-discrepancy (Sobol) sweep of 1024 points over the seven dimensionless controls of the proper multi-hop model—connectivity ρ_r, activation threshold ϕ, Lévy exponent α, patch-richness heterogeneity β, resource

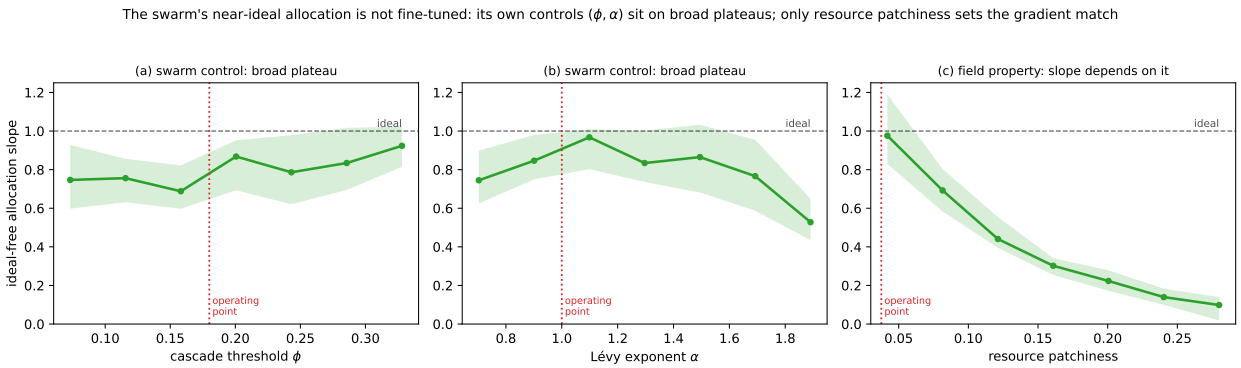


Figure 11: **The near-ideal allocation is not fine-tuned.** Ideal-free matching slope (binned median, IQR band) against three controls, from the 1024-point Sobol sweep; the dashed line is the ideal slope 1 and the dotted line marks the operating point. **(a)** The activation threshold ϕ and **(b)** the Lévy exponent α are the swarm's *own* controls: both sit on broad plateaus—the slope is near-ideal across the whole swept range of ϕ and over a wide band of α, with α=1 (inverse-square Lévy) at the peak—so neither is tuned. **(c)** Resource *patchiness* is a property of the *field*, not a swarm parameter, and it is the one axis the gradient match depends on: the slope degrades as clumps spread out, and the operating point sits in the tight-clump region where matching is near-ideal.

coverage ρ_s, resource patchiness, and the relay decay—each point evaluated over B=64 independent replica worlds, scoring the ideal-free matching slope and the fraction of the oracle ceiling. Because the box is wide, most random points fall in poor regimes; the question is whether the operating point sits on a broad, forgiving plateau in the controls the swarm itself sets, and what the allocation actually depends on.

Two findings follow (Fig. 11). First, the swarm's own controls are *not* fine-tuned: the matching slope is near-ideal across the entire swept range of the activation threshold ϕ (0.05–0.35) and over a wide band of the Lévy exponent α, with the inverse-square value α=1 sitting at the peak rather than on a cliff. This is the quantitative form of the "broad plateau" the main text relies on when it says α and ϕ are held at a-priori values and are not harvest-optimal (§4). Second, the one control the gradient match genuinely depends on is the resource *patchiness*—a property of the environment, not of the swarm: when clumps are tightly concentrated (as in the operating fields) matching is near-ideal, and it degrades as the field spreads out. This locates the scope of the result honestly: the swarm reaches the ideal-free allocation when resource is clustered, which is the common ecological case the paper targets, and not when it is diffuse. This sweep varies the swarm's *own* allocation slope across the patchiness axis; the head-to-head swarm-vs-learner *harvest* contest (§6) is measured at the two field endpoints—clustered and scarce—and is not swept across intermediate patchiness, which is why we say it *bounds* rather than settles the general comparison; a full harvest sweep against trained learners over the patchiness axis is left to future work. (The sweep holds the agent count N=200 fixed; scaling in N is a separate axis we do not vary here.)